Day 1: Foundational Nutrients & Baseline Vitals

The Paradigm Shift: From Speculation to Molecular Certainty

Most people begin a health journey by buying a cupboard of supplements: a high-dose vitamin here, an "energy" stack there. The problem is that nobody can know which nutrients their body lacks, which it has in excess, and which may interact with a medicine or condition they already have. Without a baseline blood test, every high-dose product is a guess, and guesses cost money. A person with normal ferritin who buys iron pays for a product that does nothing and may cause harm. A person already replete in vitamin D who doubles the dose gains little and risks excess calcium. Money spent on speculation is money not spent on measurement.

The biological risk is larger than the financial one. Iron accumulates when it is not needed. Fat-soluble vitamins are stored rather than excreted. Some supplements change how medicines such as blood thinners, thyroid replacement or blood-pressure drugs behave. The principle of this workbook is simple: **measure first, then target.** Your GP is your co-pilot, and this document helps you arrive with data rather than hunches.

THE "SAFETY FIRST" PRINCIPLE. While you wait for test results, establish a small, low-risk, universal nutritional foundation at conventional, label-directed doses. These four nutrients support nearly everyone, are widely tolerated, and do not pre-empt what your results might show. Therapeutic or high-dose protocols wait for laboratory evidence.

The Day 1 Universal Baseline Stack

Nutrient	Biological role and why it is in the stack
1. Vitamin D3 + K2 (MK-7)	Vitamin D3 is converted to calcitriol, which binds the nuclear vitamin D receptor and regulates genes involved in calcium absorption, bone mineralisation, and immune and muscle function. Vitamin K2 (MK-7) activates calcium-handling proteins, osteocalcin in bone and matrix Gla protein in artery walls, helping direct calcium into skeletal tissue and supporting vascular compliance (arterial flexibility). Together they support calcium homeostasis. *Note: K2 interacts with warfarin; see safety box.*
2. Omega-3 (high-concentration EPA/DHA)	EPA and DHA are built into cell membranes, where they influence membrane fluidity and receptor function. They are precursors to specialised pro-resolving mediators (resolvins, protectins) that help switch off inflammation. They also lower circulating triglycerides and support cardiovascular and brain health. Choose a product that states its EPA+DHA content per serving, not just "fish oil".
3. Magnesium glycinate / malate	Magnesium is a cofactor in over 300 enzyme reactions. It supports autonomic nervous system balance and calms excitatory NMDA receptor activity, which helps the nervous system settle. Glycinate is well tolerated and favoured for sleep onset and architecture; malate supports muscular recovery and energy metabolism. Both are gentler on the gut than oxide or citrate.
4. Bio-available daily micronutrient matrix	A well-formulated multi-nutrient supplying B-vitamins (as methylated or coenzyme forms), zinc, selenium and trace minerals, which act as cofactors in mitochondrial ATP production and baseline enzyme function. Choose one *without* high-dose iron unless a blood test shows you need it.

14-Day Baseline Vitals Protocol

- **Morning resting heart rate:** on waking, before caffeine, sit quietly for 5 minutes, then count or record your pulse for 60 seconds.
- **Blood pressure:** same time daily, seated, back supported, feet flat, arm at heart level, after 5 minutes' rest. Take two readings one minute apart and record both. Avoid caffeine and exercise for 30 minutes before.
- **Hydration:** note your water intake (glasses) and first-morning urine colour (1 = pale straw, 3 = amber, 5 = dark). Record patterns, not single days.

Day	Resting HR (bpm)	BP reading 1	BP reading 2	Water (glasses)	Urine colour (1–5)
1					
2					
3					
4					
5					
6					
7					

Days 8–14: continue on the Page 4 worksheet or a copy of this table.

⚠ CLINICIAN SAFETY NOTICE

Day 1 nutrients provide **universal support only**. They are not a treatment. Targeted therapeutic doses, including **iron, high-dose vitamin D3, B12 injections, and metabolic or hormonal activators**, must await laboratory confirmation and your doctor's advice. Tell your GP before starting any supplement if you take **blood thinners (warfarin and similar), blood-pressure or diabetes medicines**, have kidney disease, are pregnant or breastfeeding, or have a bleeding disorder. High-dose omega-3 can increase bleeding tendency; magnesium accumulates in reduced kidney function; vitamin K2 opposes warfarin. If you develop chest pain, severe breathlessness, fainting or a BP of 180/120 or above, seek urgent medical care. This document is educational and does not replace medical advice.

THE PROGRESSIVE DIAGNOSTIC ROADMAP

Level 1 vs. Level 2 Testing Framework

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LEVEL 1: The Core Tri-Marker Anchor HIGH PRIORITY

Three tests give the most information for the least cost and effort. Start here, then decide with your GP whether Level 2 is warranted.

1. ApoB (Apolipoprotein B)

Every atherogenic particle (LDL, VLDL, IDL, Lp(a)) carries exactly one ApoB molecule, so ApoB is a direct **count of the particles** that can enter and lodge in artery walls. Standard LDL-C only measures the cholesterol *mass* carried, which can look normal when particles are numerous but small, a pattern common in insulin resistance and high triglycerides. This discordance is why ApoB is a better predictor of plaque risk than LDL-C or total cholesterol, and why guidelines increasingly endorse it.

2. HbA1c (Glycated Haemoglobin)

Glucose attaches to haemoglobin in proportion to blood sugar, and red cells live about 90–120 days, so HbA1c reflects your **average glucose over roughly three months**. It identifies pre-diabetes and early insulin resistance before a fasting glucose turns abnormal. Chronic high glucose also drives formation of advanced glycation end-products (AGEs), which stiffen collagen and vessels. Anaemia and some haemoglobin variants can distort results, so interpret with your GP.

3. hs-CRP (High-Sensitivity CRP)

The liver produces C-reactive protein in response to inflammatory signals. The high-sensitivity assay detects the **low-grade, subclinical inflammation** linked to vascular risk, metabolic strain and chronic immune stress. Single values rise with infection, injury or hard exercise, so avoid testing when unwell, and repeat if a result is high (above about 3 mg/L).

LEVEL 2: Comprehensive Multi-System Panel (56 Biomarkers)

For example, the Lola Vital Check 56 or the HealthLabs Cellular Map. Ask your GP which markers are available through your health system and which need a private panel.

System	Biomarkers	What it tells you
Lipid sub-fractions	ApoB, ApoA1, Triglycerides, HDL-C, Direct LDL-C	Atherogenic particle burden against protective HDL particles; ApoB:ApoA1 ratio refines risk.
Metabolic & glycaemic	Fasting glucose, HbA1c, Fasting insulin	Short and long-term glucose control; insulin shows resistance years before glucose rises.
Inflammatory & vascular	hs-CRP, Homocysteine, Fibrinogen	Inflammation, B-vitamin methylation status, clotting tendency.
Organ function & enzymes	ALT, AST, GGT, eGFR, Creatinine, Electrolytes (Na, K, Cl)	Liver stress, kidney filtration, hydration and electrolyte balance.
Hormonal & endocrine	Total & Free Testosterone, Estradiol, DHEA-S, SHBG	Sex-hormone status, adrenal reserve, and how much hormone is bioavailable.
Thyroid cascade	TSH, Free T3, Free T4	Pituitary signal, active hormone, and storage hormone; metabolic rate and energy.
Micronutrients & storage	Vitamin D3 (25-OH), B12, Serum iron, Ferritin, Folate	Confirms whether the Day 1 stack is needed, and whether iron or B12 dosing is justified.

The seven rows above list the key markers; full kits add related full blood count, lipid and mineral values to reach 56.

Your Progressive Pathway

✓	Step	Date booked	Result received
☐	Level 1: ApoB, HbA1c, hs-CRP		
☐	Review results with GP; decide on Level 2		
☐	Level 2: Full 56-biomarker panel		
☐	Targeted protocol agreed with GP		

TESTING TIPS Fast 8–12 hours (water allowed) for glucose, insulin and triglycerides. Test in the morning. Pause biotin supplements 48 hours before, as it can distort some immunoassays (including thyroid), and tell the phlebotomist about all medicines. Reference ranges vary by lab; always interpret with a clinician.

ANNUAL GP PHYSICAL EXAM & CONTINUOUS BIO-TELEMETRY

In-Person Physical Diagnostics & Hardware Telemetry

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Section A: The Annual GP Physical Examination Protocol

Blood tests cannot replace a hands-on examination. Use this checklist to understand what a thorough check involves and to tick off what your GP covers.

✓	Examination	What your GP is assessing
☐	**Cardiopulmonary auscultation**	Stethoscope check of the S1 (“lub”, valve closure at the start of contraction) and S2 (“dub”) heart sounds, rhythm regularity, and any murmurs, extra sounds or irregular beats. Lung fields are listened to front and back for wheeze, crackles or reduced air entry.
☐	**Peripheral vascular assessment**	Pulses at the wrist, groin, behind the knee and foot; skin temperature and colour; capillary refill; ankle swelling; carotid check for bruits.
☐	**Resting blood pressure**	Properly cuffed, seated, ideally repeated, and compared against your 14-day home readings to spot white-coat or masked hypertension.
☐	**Abdominal palpation**	Gentle assessment of tenderness, organ enlargement (liver, spleen), masses or aortic pulsation.
☐	**Lymph node assessment**	Neck, armpit and groin nodes checked for size, tenderness and mobility.

Section B: 14-Day Continuous Bio-Telemetry (Hardware Integration)

Wearable vitals

Heart rate variability (HRV) is the beat-to-beat variation that reflects vagal (parasympathetic) tone and recovery capacity. Compare *your own* trend rather than other people’s numbers. **Resting HR** reveals fitness and stress load. **Sleep architecture** records light, deep and REM proportions; consumer devices are estimates, not clinical sleep studies.

Segmental body composition

A multi-frequency BIA scale passes a safe low-level current through the body to estimate **visceral fat**, **muscle mass distribution** by limb and trunk, and **hydration balance**. Weigh at the same time under the same conditions (morning, fasted, after the bathroom), as hydration shifts readings.

Continuous glucose monitoring (CGM)

A small skin sensor records glucose every few minutes, mapping **postprandial spikes** and **glycaemic variability** (how widely glucose swings). Useful alongside HbA1c to learn which meals suit you. Discuss with your GP, especially if you have symptoms.

Metric to log (14 days)	Week 1 average	Week 2 average	Trend (↑ ↓ →)
HRV (ms)			
Resting HR (bpm)			
Deep sleep / REM (min)			
Visceral fat score			
Muscle mass (kg)			
Peak post-meal glucose (mmol/L)			

Section C: Epigenetic Cellular Age Testing

Chronological age counts birthdays; **biological age** estimates how your cells are actually ageing. Whole-methylome gDNA testing reads DNA methylation, the chemical tags on your DNA that switch genes on and off, at hundreds of thousands of sites, then compares your pattern with population reference data to estimate a biological age. Because methylation responds to sleep, nutrition, movement and stress, it can serve as a long-term benchmark: test at baseline, then repeat no sooner than every 6–12 months. Results are research-grade estimates, not a diagnosis, so use them as a trend and not a verdict.

INTERPRETING TELEMETRY Wearable and CGM data are indicative, not diagnostic. Bring trends and questions to your GP rather than acting on single readings. Persistent chest symptoms, palpitations, fainting or a consistently high resting HR warrant medical review regardless of device data.

PATIENT-HELD CLINICAL RECORD & CO-EVALUATION WORKSHEET

Printable GP Consultation Template & Doctor Script

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Patient name _____ GP name _____

Date of birth _____ Clinic name _____

Date _____

Block 1: 14-Day Wearable Vitals & Symptom Summary

Measure	Average	Lowest	Highest	Notes
Resting HR (bpm)				
HRV (ms)				
Blood pressure (mmHg)				
Sleep (hours)				
Key symptoms (fatigue, pain, mood…)				

Block 2: Current Lifestyle & Day 1 Supplement Log

☐ Vitamin D3 + K2 dose: _____

☐ Omega-3 EPA/DHA dose: _____

☐ Magnesium glycinate/malate dose: _____

☐ Micronutrient matrix dose: _____

Started on: _____

Current medicines & other supplements:

_____

_____

_____

Allergies: _____

Block 3: Requisition Checklist for GP

☐ **ApoB** (Level 1)

☐ **HbA1c** (Level 1)

☐ **hs-CRP** (Level 1)

☐ **Full 56-biomarker requisition** (Level 2)

☐ Vitamin D, B12, ferritin (if not in panel)

☐ Other: _____

Block 4: Collaborative Doctor Discussion Script

1. Open with partnership. “I’ve been tracking my blood pressure, heart rate and sleep for two weeks, and I’d like to understand my baseline. Could we look at this data together?”

2. Ask for the evidence-based tests. “I’ve read that ApoB, HbA1c and hs-CRP give a clearer picture of cardiovascular and metabolic risk than a standard cholesterol check. Would these be appropriate for me, and if not, what would you recommend instead?”

3. Agree the plan. “Whatever you decide, could you explain what would change your advice, how often we should retest, and whether my supplements are safe alongside my medicines?”

Block 5: Clinician Directive

CLINICIAN NOTES & COLLABORATIVE CARE DIRECTIVES

_____

_____

_____

_____

_____

_____

Clinician signature _____ Print name _____ Date _____